

SYNOPSIS

At 0154 on 11 January 2014, the Liberia registered multi-purpose cargo ship *Rickmers Dubai* collided with *Walcon Wizard*, an un-manned crane barge which was being towed by the tug *Kingston* in the south-west traffic lane of the Dover Strait Traffic Separation Scheme. Following the collision, *Kingston* was pulled stern-first through the water until the towline ran free from its tow winch. *Walcon Wizard* was badly damaged and *Rickmers Dubai*'s hull was punctured above the waterline. The hydraulic system for the tow winch on board *Kingston* was also damaged. There were no injuries and there was no pollution. The accident occurred while the cargo ship was overtaking the tug and tow.

The investigation established that *Rickmers Dubai*'s officer of the watch:

- Had not kept a proper lookout. He did not see *Walcon Wizard* when he altered course to avoid *Kingston*, which was less than 2 cables ahead.
- Relied solely on AIS information displayed on the ECDIS as an aid to collision avoidance.
- Was relatively inactive during his watch. He was alone on the bridge, he did not monitor the radar and the bridge navigational watch alarm system was switched off.
- Did not take note of the content of two safety broadcasts issued by Dover Coastguard advising of *Kingston* and *Walcon Wizard*'s position in the south-west traffic lane.

The investigation also identified that:

- Neither *Kingston* nor *Walcon Wizard* were transmitting on AIS.
- *Kingston*'s towing and stern lights were probably obscured by a floodlight.
- A safety broadcast, which was scheduled for 0140 and would have included the position and movement of *Kingston* and *Walcon Wizard*, was not issued by Dover Coastguard because the operator was busy dealing with other matters and had been on watch for almost 4 hours.
- The towline was secured to the tow winch on board *Kingston* and could not be released quickly or from the wheelhouse.
- The requirements for the release of towlines from towing winches are ambiguous.

Recommendations have been made to the ships' managers designed to improve the navigational and operational safety of their fleets. A recommendation has also been made to the Maritime and Coastguard Agency that is intended to improve the effectiveness of the safety messages broadcast by Dover Coastguard.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF *RICKMERS DUBAI*, *KINGSTON* & *WALCON WIZARD* AND ACCIDENT

SHIP PARTICULARS	
Vessel's name	<i>Rickmers Dubai</i>
Flag	Liberia
Classification society	Germanischer Lloyd
IMO number/fishing numbers	9467134
Type	Multi-purpose dry cargo
Registered owner	Kg Schifffahrtsgesellschaft MS Baltic Winter mbh &Co.
Manager(s)	Reedereiverwaltung Heino Winter GmbH & Co.KG
Construction	Steel
Year of build	2011
Length overall	166.15m
Length (bp)	156m
Gross tonnage	15377
Minimum safe manning	10
Authorised cargo	Dry, containerised & heavy cargo
VOYAGE PARTICULARS	
Port of departure	Hamburg
Port of arrival	Genoa
Type of voyage	Short international
Cargo information	General cargo
Manning	17

MARINE CASUALTY INFORMATION	
Date and time	11 January 2014 at 0154
Type of marine casualty or incident	Serious Marine Casualty
Location of incident	South-west lane of Dover Strait Traffic Separation Scheme
Place on board	Bow
Injuries/fatalities	None
Damage/environmental impact	Hull punctures and minor indentations
Ship operation	On passage
Voyage segment	Mid-water
External & internal environment	Wind: south-westerly wind force 5 to 6 Sea state: moderate to rough Visibility: good
Persons on board	17



Rickmers Dubai

SHIP PARTICULARS		
Vessel's name	Kingston	Walcon Wizard
Flag	UK	UK
Classification society	Not applicable	Not applicable
IMO number	5344437	908792
Type	Tug	Crane barge
Registered owner	Griffin Towage & Marine	Walcon Marine Ltd
Manager(s)	Griffin Towage & Marine	Walcon Marine Ltd
Construction	Steel	Steel
Year of build	1961	2003
Length overall	27.2m	23.99m
Length bp	25.3m	Not applicable
Gross tonnage	113	105.6
Minimum safe manning	Not applicable	Not applicable
Authorised cargo	None	165t deck cargo
VOYAGE PARTICULARS		
Port of departure	Ipswich	
Port of arrival	Southampton	
Type of voyage	Coastal	
Cargo information	Not applicable	
Manning	4	Unmanned
MARINE CASUALTY INFORMATION		
Date and time	11 January 2014 at 0154	
Type of marine casualty or incident	Serious Marine Casualty	
Location of incident	South-west lane of Dover Traffic Separation Scheme	
Place on board	Deck	Superstructure
Injuries/fatalities	None	
Damage/environmental impact	Tow winch hydraulics and brake damaged	Substantial damage to wheelhouse and crane
Ship operation	Towing	Under tow
Voyage segment	Mid-water	
External & internal environment	Wind: south-westerly wind force 5 to 6 Sea state: moderate to rough Visibility: good	
Persons on board	4	None

1.2 NARRATIVE

1.2.1 Events leading to the collision

At 0115¹ on 11 January 2014, the multi-purpose cargo ship *Rickmers Dubai* was transiting the south-west lane of the Dover Strait Traffic Separation Scheme (TSS). The vessel was on passage from Hamburg, Germany, to Genoa, Italy, with cargo in its hold and on deck. *Rickmers Dubai* was making good a course over the ground (COG) of 233° at a speed over the ground (SOG) of 15.4 knots (kts) (**Figure 1**).



Figure 1: CNIS display at 0115

Directly ahead of *Rickmers Dubai*, at a range of 6.3nm was the tug *Kingston*, which was towing the unmanned crane barge *Walcon Wizard*. The tug and tow were on passage from Ipswich, UK to Southampton, UK. The length of the wire towline was 250m. The tug was making good a COG of 237° at a speed of 6kts and was keeping to the northern edge of the traffic lane to enable other vessels to pass down its port side. The predicted closest point of approach (CPA) between *Rickmers Dubai* and *Kingston* was about 250m at 0155.

The wind was from the south-west between Beaufort force 5 and 6. The sea was moderate to rough and the visibility was good. The predicted tidal stream was setting to the south-west at a rate of 1kt.

At 0145, the distance between *Rickmers Dubai* and *Kingston* had reduced to 1.6nm and the CPA of the vessels was 40m (**Figure 2**). At about this time, *Kingston*'s wheelhouse watchkeeper became aware of the cargo ship approaching on the tug's starboard quarter. He assessed that, although the cargo ship was close, it would pass clear down the starboard side of *Walcon Wizard* and *Kingston*. The watchkeeper noticed that the automatic identification system (AIS) data included the

¹ The local time kept on board *Rickmers Dubai* was UTC (+1), therefore ship's time was 0215.



Figure 2: CNIS display at 0145

cargo ship's maritime mobile service identity (MMSI) number, but not its name. He started to note the MMSI number on a piece of paper in case he needed to contact the cargo ship by radio.

Shortly after, *Rickmers Dubai's* second officer, who was the officer of the watch (OOW), saw *Kingston's* lights very fine off the cargo ship's port bow. At 0154, he adjusted the autopilot to port in order to overtake the tug on its port side (Figure 3). Seconds later, *Rickmers Dubai* collided with *Walcon Wizard*. It then passed between the crane barge and the tug, catching on the towline as it did so.

Kingston's mate, who was the tug's wheelhouse watchkeeper, saw that the cargo ship was now passing very close to the tug's stern. At 0155, he broadcast on very high frequency (VHF) radio channel 16:

"What are doing? Slow down, there is a wire, slow down, slow down!"

The VHF call was heard by *Rickmers Dubai's* OOW, who quickly reduced the ahead pitch on the controllable pitch propeller (CPP) from 83% to 29%. As *Rickmers Dubai* overtook *Kingston*, the lead of the towline moved from astern to ahead, which pulled the tug through 180°. *Kingston* was then towed stern-first.

Kingston's mate immediately took the main engine out of gear and shouted down to the accommodation to warn the rest of the tug's crew. The tug's skipper and engineer immediately went on deck and saw that the towline was being pulled off the tow winch drum, dense smoke was coming from the winch drum brake, the aft deck was under water and hydraulic oil was spraying from ruptured connections. Seconds later, the bitter end of the wire towline broke free from the drum. *Kingston* then stopped in the water with *Walcon Wizard* off its starboard side.



Figure 3: CNIS display at 0154

1.2.2 Subsequent actions

Kingston and Walcon Wizard

At 0159, *Kingston*'s watchkeeper informed Dover Coastguard of the collision and that the tug's tow had been lost. He also informed the coastguard that the tug's crew were uninjured. Dover Coastguard alerted the Dungeness all weather lifeboat (ALB), which was launched to assist. It also advised approaching vessels to give *Kingston* and *Walcon Wizard* a wide berth.

Kingston was manoeuvred close to *Walcon Wizard* but its crew were unable to access the emergency towline, which had been prepared on the barge before leaving Ipswich, because of damage to the barge's crane jib. The damage to the tug's hydraulic system also prevented the crew from recovering the lost towline, which was anchoring *Walcon Wizard* to the seabed.

At 0234, the Dungeness ALB arrived on scene. Its crew confirmed that, except for the stern light, *Walcon Wizard*'s navigation lights were clearly visible. The barge's stern light was illuminated but it was obscured by damage to its superstructure.

The towline was eventually cut free from *Walcon Wizard* by the crew of *THV Galatea*, which then took the crane barge in tow. *Kingston* and *Walcon Wizard* berthed alongside in Dover at 2300 on the same day.

Rickmers Dubai

At 0157, *Rickmers Dubai*'s chief engineer, having been awoken by a change in engine noise and increased ship's vibration, arrived on the bridge. At 0202, Dover Coastguard called *Rickmers Dubai* by VHF radio. The second officer responded immediately, stating:

"The tug was not showing any signal."

Dover Coastguard requested the second officer ensured that the voyage data recorder (VDR) was saved and that details of the incident were entered in the deck logbook. By 0205, *Rickmers Dubai*'s speed had reduced to 6kts as the vessel continued on a south-westerly heading.

At 0212, the second officer started to slowly increase the vessel's speed. The chief engineer prompted the OOW to inform the master of what had happened. He then left the bridge.

At 0217, the second officer informed Dover Coastguard that *Kingston* did not, and was still not, transmitting on AIS. He also stated that he had not seen the towline or spoken to the tug before overtaking. The OOW also suggested the towed barge was not showing navigation lights.

By 0235, *Rickmers Dubai*'s speed had increased to 13.5kts. At about 0300, the chief officer arrived on the bridge to take over the navigational watch. During the watch handover, the second officer informed the chief officer that *Rickmers Dubai* had passed between a tug and its tow, but that the cargo ship had not been damaged. On completion of the watch handover, the chief officer instructed the second officer to make sure that there was no damage. Accordingly, the second officer went onto the forecastle with a torch and looked over the bow. No damage was seen.

At 0330, Dover Coastguard called on VHF radio and requested a damage report. The chief officer replied that he would check and respond within 30 minutes.

At 0356, Dover Coastguard instructed *Rickmers Dubai* to leave the TSS, anchor and await further instructions. The chief officer called the master to the bridge. The chief officer then briefed the master, who immediately called the bosun and instructed him to sound the ballast tanks. No unusual soundings were noted.

At 0742 *Rickmers Dubai* anchored off Newhaven.

1.3 DAMAGE

1.3.1 Tug and tow

Walcon Wizard's wheelhouse was set forward in the collision and crushed. The jib of its crane was buckled and guardrails were also bent (**Figure 4**). *Kingston*'s hydraulic motor for the tow winch was seriously damaged and several hydraulic connections were ruptured.

1.4.4 Voyage data recorder

Information recorded by *Rickmers Dubai*'s VDR was saved when the vessel was at anchor off Newhaven. Review of the recorded data identified, amongst other things, that:

- The X-band radar display was set to the 6nm range scale and the origin was offset to the north east in order to extend the view ahead. By 0100, the radar targets of both *Kingston* and *Walcon Wizard* were visible at a range of 10nm (**Figure 9**).
- The radar targets associated with *Kingston* and *Walcon Wizard* merged between 0147 and 0150 when at a range of about 1nm (**Figure 10**).
- A radar alarm of 5 cables/1 minute was set on the ARPA for operator selected targets only (**Figures 9 and 10**).
- *Thun Goliath* was the only AIS target selected after midnight (**Figures 9 and 10**).
- No radar targets were acquired using ARPA.
- The vessel had been on a steady course and speed since 0100 and was approximately 3 cables to the north of its intended track.
- There was no crossing traffic in close proximity.
- The VHF radios were set to channels 11 and 16. Safety broadcasts transmitted by Dover Coastguard were loud and clear.
- No English was spoken on the bridge during the second officer's watch.
- There was no evidence of persons entering or leaving the bridge during the hour before the collision.
- Recorded music was being played throughout on the computer sited on the port side of the bridge (**Figure 7**).

The VDR was saved by the vessel's crew to a Universal Serial Bus (USB) dongle, which was bright orange and labelled as 'only for use with the VDR'. When MAIB inspectors cloned the USB dongle, they identified that a movie had been deleted.

1.4.5 Safety management system and procedures

Rickmers Dubai's safety management certificate (SMC) was issued by Germanischer Lloyd on behalf of the Republic of Liberia, following an initial ISM audit on 17 January 2012. The audit identified three non-conformities and made three observations, none of which were related to navigation or bridge watchkeeping.

1.7 COLLISION REGULATIONS

The following summarised rules from the International Regulations for the Prevention of Collisions at Sea 1972 (COLREGS) are particularly relevant to this accident:

Rule 5 - Look-out

“Every vessel shall at all times maintain a proper look-out by sight and hearing as well as by all available means appropriate in the prevailing circumstances and conditions to make a full appraisal of the situation and risk of collision.”

Rule 10 - Traffic Separation Schemes. This rule specifies the responsibilities between vessels operating in a traffic separation scheme including:

“(d) (i) A vessel shall not use an inshore traffic zone when she can safely use the appropriate traffic lane within the adjacent traffic separation scheme. However, vessels less than 20 metres in length, sailing vessels and vessels engaged in fishing may use the inshore traffic zone.”

Rule 13 - Overtaking. This rule states that the overtaking vessel must keep out of the way of the vessel being overtaken.

Rule 24 – Towing and Pushing. This rule requires that a power-driven vessel when towing shall exhibit: (i) instead of the light prescribed in Rule 23 (a)(i) or (a)(ii), two masthead lights in a vertical line. When the length of the tow, measuring from the stern of the towing vessel to the after end of the tow exceeds 200 meters, three such lights in a vertical line; (ii) sidelights; (iii) a sternlight; (iv) a towing light in a vertical line above the stern light.

The rule also requires that a vessel or object being towed, other than those mentioned in paragraph (g) of this Rule, shall exhibit: (i) sidelights; (ii) a sternlight. The required visibility of these lights is 2nm (Rule 22).

Rule 27(c) – This rule requires that a power-driven vessel engaged in towing operations such as severely restricts the towing vessel and her tow in their ability to deviate from their course shall display RAM lights and shapes.

1.8 THE CARRIAGE AND USE OF AIS

1.8.1 Carriage requirements

The International Convention for the Safety of Life at Sea 1974, as amended (SOLAS), requires all cargo ships of 300gt and over which are engaged on international voyages to be fitted with AIS. European Union (EU) Directive 2002/59/EC extended this requirement to all cargo ships greater than 300gt trading domestically as well as high speed craft. Directive 2009/17/EC, amending Directive 2002/59/EC, establishing a Community vessel traffic monitoring and information system, requires European Community fishing vessels of more than 15m length overall to operate an AIS at all times.

1.13 SIMILAR ACCIDENTS

1.13.1 Dover Strait TSS – south-west traffic lane

The number of collisions and hazardous incidents (that resulted in a HAZREP report being issued)⁸ in the south-west traffic lane of the Dover Strait TSS recorded by Dover Coastguard between 2009 and 2013 is at **Table 1**.

	2009	2010	2011	2012	2013
Hazardous Incident Reports	49	54	39	28	29
Collisions	0	1	2	1	2
Total of reporting vessels	39096	39125	39074	36804	35551

Table 1 – Collisions and hazardous incidents in the south-west traffic lane of the Dover TSS (Source: Dover Coastguard 2014)

The collision data has been collected from incidents involving merchant and/or fishing vessels. The total volume of reporting vessels includes those vessels that are required to observe the CALDOVREP mandatory reporting system. The figures do not include fishing vessels and yachts that due to their size are not required to report to Dover Coastguard.

1.13.2 Collisions while overtaking in the Dover Strait

The following collisions occurred in the Dover Strait TSS and were the subject of MAIB investigation reports.

- *Eastfern and Kinsale*

On 25 September 2000, the bulk carrier *Kinsale* collided with the general cargo ship *Eastfern* in the south-west traffic lane of the Dover Strait TSS. *Kinsale* was the overtaking vessel and its OOW was preoccupied with other duties. He failed to see *Eastfern* ahead. The bridge lookout had left the bridge 20 minutes before the collision. (MAIB investigation report 18/2001.)

- *Ash and Dutch Aquamarine*

On 9 October 2001, the chemical tanker *Dutch Aquamarine* collided with the general cargo vessel *Ash* in the south-west traffic lane of the Dover Strait TSS. *Ash* was holed in the collision, listed quickly to starboard, capsized and sank. The master of *Ash* drowned. Although the weather was fine and the visibility good, the OOW on *Dutch Aquamarine* did not notice *Ash* either visually or on radar until it was too late to avoid a collision. Although the OOW on board *Ash* had seen *Dutch Aquamarine* approaching from astern he was

⁸ A HAZREP is a notification of an apparent breach of a COLREG other than rule 10. The data in the table represents both reported and unreported incidents. A reported incident is where a vessel makes a complaint about the conduct of another. An unreported incident is one in which two vessels are involved in an apparent close quarter situation where neither vessel makes a complaint about the other.

distracted by a mobile phone call minutes before the collision. *Ash* had been overtaken at a distance of only 0.1nm previously so the OOW assumed that *Dutch Aquamarine* might do likewise. (MAIB investigation report 7/2003.)

- *Spring Bok* and *Gas Arctic*

On 24 March 2012, the general refrigerated cargo ship *Spring Bok* collided with the liquefied petroleum gas tanker *Gas Arctic* in the south-west traffic lane of the Dover Strait TSS in visibility of less than 2nm. There were no injuries or pollution, but both vessels suffered structural damage. Although both OOWs had detected and identified the other vessel by radar and AIS, neither OOW made a full appraisal of the risk of collision, nor took appropriate action to prevent the collision. Even though both vessels' SMSs required additional safety precautions in visibility less than 3nm, neither posted a lookout nor sounded signals. Cumulative fatigue was also considered a contributory factor in the decision making of the OOW on *Spring Bok*. (MAIB investigation report 24/2012.)

1.13.3 Collisions involving tug and tows

The following case was the subject of a preliminary examination by the MAIB in 2006 and involved a fishing vessel colliding with a tug and tow.

- *Natalie* and *Bay Protector*

On 12 August 2006, the fishing vessel *Natalie* collided with an unlit barge that was being towed by the tug *Bay Protector*. The barge lights had failed and the tug did not issue any warnings via VHF radio. Neither of the skippers involved used all the available means to assess if there was a risk of collision.

1.13.4 Tug capsize accidents

The following MAIB investigations of the capsize of tugs while towing identified safety issues regarding the use and/or effectiveness of towline emergency release mechanisms.

- *Flying Phantom*

On 19 December 2007, *Flying Phantom* girted and sank with the loss of three crew while operating as a bow tug in thick fog. Although activated, the tow winch emergency release was slow to operate, resulting in the tug capsizing and the tow rope parting. As the engine room watertight door was left open, rapid downflooding followed. The investigation also identified that crew training and the procedures to operate in the conditions encountered were inadequate. Recommendations were made to CAs to develop a standard for towline release systems and to ensure that the systems were tested. (MAIB investigation report 17/2008.)

- *Ijsselstroom*

On 14 June 2009, *Ijsselstroom* was operating as a stern tug, running astern, with a single wire connected over the stern to a large barge. The tug skipper had intended to maintain his position and heading by using differential thrust on his two engines. As the tug towing the barge increased speed, *Ijsselstroom* yawed uncontrollably. The tug sheered, girted and then sank. The tug's crew were rescued. It was identified that the tow speed was too fast, no gob rope or bridle was in use to prevent the tug girting, and that the towline emergency release system was not operated. (MAIB investigation report 4/2010.)

- *Chiefton*

On 12 August 2011, *Chiefton* capsized and foundered following a collision with a crane barge it was towing on the River Thames. One crewman died. The inability to lengthen the tow placed *Chiefton* in danger when transiting at high speed. Also, the tow hook emergency release was not used and, even if it had been, it would probably have been ineffective. Poor watertight integrity allowed rapid downflooding, which led to the loss of the tug.

A recommendation to the MCA resulted in the issue of a surveyor's advice note (SAN) 57 in December 2013 which was aimed at ensuring that surveyors' checklists included a test of the emergency release system from all operating positions (**Annex H**). (MAIB investigation report 12/2012.)

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 THE COLLISION

As *Rickmers Dubai*'s heading was altered to port at 0154, in order to cross under *Kingston*'s stern, *Walcon Wizard* was very close off the cargo ship's port bow (**Figure 19**). The cargo ship's port bow struck the crane jib protruding from the stern of *Walcon Wizard*, causing the jib to buckle. The cargo ship's port bow then made heavy contact with the barge's superstructure causing considerable damage (**Figure 4**). As *Walcon Wizard* passed down the port side of *Rickmers Dubai*, the cargo ship's bow snagged the towline between the barge and *Kingston* (**Figure 20**).

On board *Kingston*, the lead of the towline quickly changed from astern to ahead, as *Rickmers Dubai* passed by. The gob rope prevented the tug from girting but *Kingston* was pulled through 180° and was then towed stern-first at 14kts, almost the same speed as the cargo ship.

At such a speed, it is not surprising that the tug's aft deck was quickly awash and that the tow winch brake started to render. As the wire was pulled from the tow winch, the resulting back pressure in the winch's hydraulic system caused joints to rupture. The smoke on the deck was caused by the tow winch brake overheating.

After the end of the towline had been pulled completely from the tow winch drum, it dropped to the seabed. *Rickmers Dubai* sailed on, leaving *Walcon Wizard* anchored by the wire towline with *Kingston* stopped in the water heading towards the north-east. In the circumstances, it was extremely fortunate that *Kingston* did not capsize and that none of its crew were injured.

2.3 RICKMERS DUBAI

2.3.1 Lookout

Before the collision, the radar targets associated with *Kingston* and *Walcon Wizard* were clearly painting on the X-band radar display on *Rickmers Dubai*'s bridge for nearly 55 minutes (**Figure 9**). Also, as the cargo ship was closing the tug and tow with a speed advantage of approximately 9kts (**Figures 1, 2 and 3**), and the range of the lights fitted on board *Walcon Wizard* was 3nm, the aft lights on the tow would have been visible from *Rickmers Dubai* for at least 20 minutes before the collision. Therefore, *Rickmers Dubai*'s OOW had ample time to detect, assess and take avoiding action. However, it is evident from his alteration to port when only 2 cables from the tug, and his failure to see *Walcon Wizard* at all, that the OOW had not been keeping a proper lookout and had only seen the tug just before he altered. Although at the time of the alteration the barge's lights were possibly obscured by the port side deck cranes (**Figure 8**), they would have been clearly visible beforehand despite the moderate to rough seas.

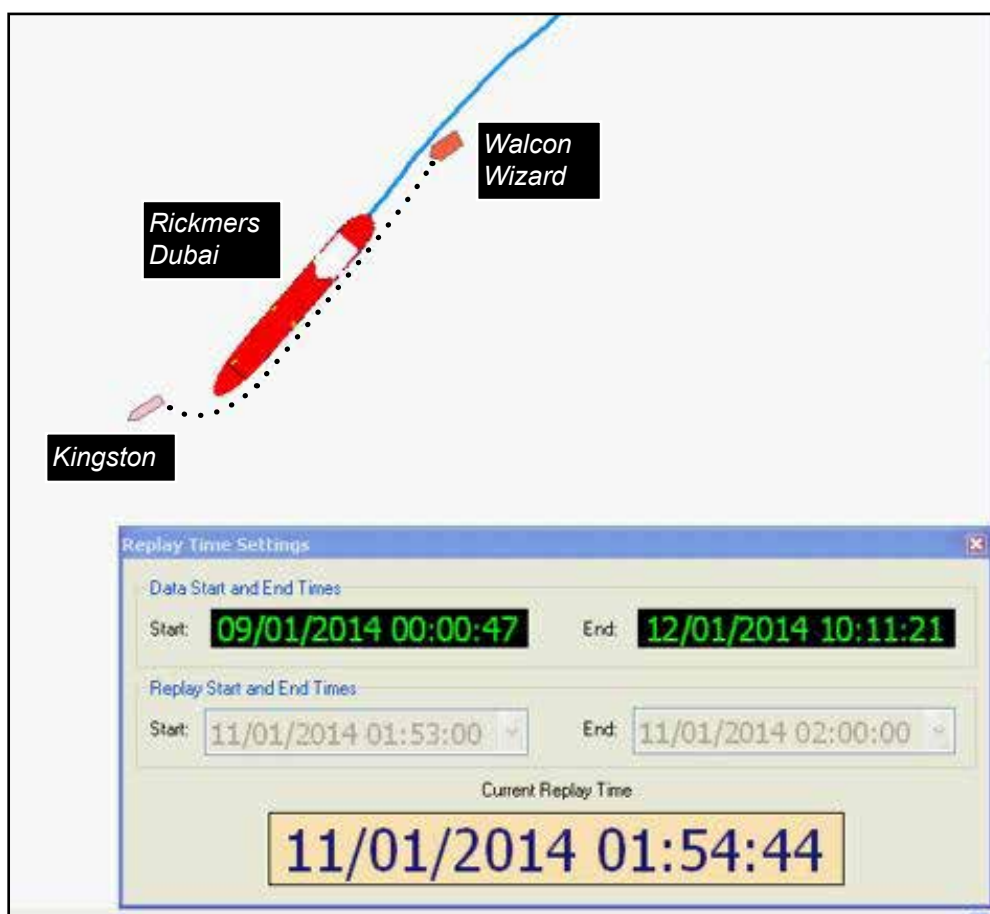


Figure 20: Simulation of the situation at 0154:44

2.3.2 Visual assessment

When *Rickmers Dubai*'s OOW saw *Kingston*'s lights ahead, it is evident that he did not associate the tug's lights with a vessel engaged in towing. This was possibly because although *Kingston* was showing a towing light above its stern light, these lights were potentially difficult to see due to the glare from the floodlight sited between them (**Figure 16**). In addition, the three masthead lights indicating a tow over 200m in length would not have been visible from astern⁹. Consequently, as the tug was also not displaying lights to indicate that it was restricted in its ability to manoeuvre, and the OOW did not see *Walcon Wizard*, his avoiding action was based on his assumption that *Rickmers Dubai* was overtaking a single vessel, not a tug and its tow.

2.3.3 Reliance on ECDIS and AIS

In view of *Rickmers Dubai*'s OOW's comments to Dover Coastguard following the collision (**paragraph 1.2.2**), which highlighted that *Kingston* had not been transmitting on AIS, it is almost certain that his late detection of *Kingston* and his ignorance of the proximity of *Walcon Wizard* were due to his reliance on AIS information shown on the ECDIS.

⁹ A masthead light is required to be visible from right ahead to 22.5° abaft the beam on either side of the vessel (namely an arc of 225°)

ECDIS is capable of providing a wealth of information to the user, including charts, waypoints, safe water and overlaid AIS information. However, it is not a 'one-stop shop' which is able to provide all of the information required by an OOW. ECDIS must be used in conjunction with other aids to navigation and collision avoidance, particularly radar and visual lookout.

In this case, the OOW had set up the X-band radar display at the start of his watch. However, VDR evidence (**paragraph 1.4.4** and **Figures 9** and **10**) indicates that the OOW did not see the radar targets of *Kingston* and *Walcon Wizard*, or that he saw them but did not acquire them using ARPA. The absence of AIS information for the radar targets associated with *Kingston* and *Walcon Wizard* should have prompted the OOW to use ARPA to determine if a risk of collision existed. The detection of radar targets directly ahead and closing should also have prompted the OOW to look out of the window and attempt to correlate the targets with visual information. As the OOW took neither of these actions, and the targets were on the X-band display for almost 1 hour, it is likely that the OOW was not monitoring the radar display at all.

The guidance issued by the IMO and the MCA regarding the use of AIS in collision avoidance (**Paragraph 1.8.2**) highlights its advantages and disadvantages, and a balance needs to be struck between over-reliance and effective use. To achieve such a balance, it is important that OOWs are fully aware of the system's capabilities and limitations. In this case, the second officer's reliance on the AIS information displayed on ECDIS for collision avoidance strongly indicates that he was not aware that many vessels, such as small fishing vessels, leisure craft, warships and vessels under 300gt might not be displayed.

2.3.4 OOW stimulation

Until seeing *Kingston*, *Rickmers Dubai*'s second officer's bridge watch had been quiet and uneventful. The ship was near to its intended track and the adjacent vessels were all heading in the same direction (**Figures 1, 2, 3**). Other than selecting the AIS target of *Thun Goliath* on the X-band radar display, there is no evidence of the OOW's other activities. In short, he appears to have been largely inactive until he altered course to port to avoid *Kingston*. The music playing on the bridge throughout the watch, the ease of monitoring the information available on the ECDIS, the lack of a periodic alarm from the BNWAS, and the absence of the watchman would all have contributed to the OOW's 'relaxed' state.

Although the OOW was well rested and did not feel tired, research has shown that alertness and performance tend to be at their lowest during the early hours of the morning. The human circadian rhythm is synchronised with the normal pattern of daytime wakefulness and sleep at night. Therefore, given that the collision occurred at 0154 (0254 ship's time), it is likely that, although the OOW might not have fallen asleep, his level of arousal was low. Consequently, the second officer was not proactive in maintaining his situational awareness or reactive to changing circumstances. This is supported by his failure to use radar or ARPA and to keep an effective visual lookout.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. *Rickmers Dubai's* OOW was not keeping a proper lookout and had only seen *Kingston* just before he altered course towards *Walcon Wizard*. [2.3.1]
2. *Rickmers Dubai's* OOW did not see *Walcon Wizard* or recognise that *Kingston* was engaged in towing. [2.3.2]
3. It is almost certain that the late detection of *Kingston* by *Rickmers Dubai's* OOW and his ignorance of the proximity of *Walcon Wizard* were due to an over-reliance on AIS information shown on the ECDIS. [2.3.3]
4. The radar targets of *Kingston* and *Walcon Wizard* were on the X-band radar display on board *Rickmers Dubai* for almost 1 hour, but the OOW did not use ARPA to determine if a risk of collision existed, or look out of the window to try and correlate the targets with visual sightings. Therefore, it is likely that he was not monitoring the radar display at all. [2.3.3]
5. It is likely that *Rickmers Dubai's* OOW's level of arousal was low. Consequently, he was not proactive in maintaining his situational awareness or reactive to changing circumstances. [2.3.4]
6. *Rickmers Dubai's* OOW was on the bridge by himself at the time of the collision despite the vessel being in the Dover Strait at night. [2.3.5]
7. No dedicated lookout was posted on the bridge of *Rickmers Dubai* despite an AB being rostered for the purpose. [2.3.5]
8. Neither *Kingston* nor *Walcon Wizard* were transmitting on AIS, which would have increased the probability of them being noticed at an early stage by *Rickmers Dubai's* OOW. [2.4.1]
9. *Kingston's* towing light and stern light were potentially obscured by the glare of a floodlight. [2.4.2]
10. The lack of a safety culture among *Rickmers Dubai's* bridge watchkeeping officers was demonstrated by a disconnection between their behaviour and the ship manager's intentions on how the vessel was to be run. [2.6.1]
11. A more considered approach to the provision of equipment and instructions on board *Kingston* might have helped to prevent *Rickmers Dubai* colliding with *Walcon Wizard*, reduce the risk of capsizing and minimise the damage on board. [2.6.2]

3.2 OTHER SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT

1. A balance needs to be struck between the over-reliance on AIS and its effective use. Therefore, it is important that OOWs are fully aware of the system's capabilities and limitations. [2.3.3]
2. The effective audit of bridge watchkeeping practices is a problem faced by all ship managers, which is not easily solved. [2.6]

3.3 SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The procedure for releasing the towline from the winch on board *Kingston* was unsuitable for use in all operating conditions as required by the SCV Code. It was certainly not suitable for use in an emergency in the dark and in rough seas. [2.5]
2. The circumstances of this accident, together with others that have resulted in the capsize of tugs, indicate there is a strong case for the emergency release arrangements for towing gear to be more clearly articulated at the earliest opportunity. [2.5]
3. A safety broadcast, which should have been issued at 0140, was not issued because the CNIS operator was busy and he had been at the CNIS desk for almost 4 hours. [2.7]
4. The inclusion in the CNIS safety broadcasts that the tug and tow was not transmitting on AIS would have made the broadcasts more informative and might have assisted in attracting the attention of some OOWs. [2.7]
5. The RAM lights on board *Kingston* were not available for use because the white light was defective. [2.4.2]

3.4 OTHER SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT¹²

1. The response of *Rickmers Dubai*'s crew following the collision was slow and lacked rigour. [2.3.6]
2. The damage to *Rickmers Dubai* was not identified until MAIB inspectors boarded the vessel several hours after the collision as the crew had been unable to launch the rescue boat and they had not anticipated damage so far above the waterline. [2.3.6]
3. The USB dongle, which should only have been used with the VDR, had been used to store a movie. Such abuse of equipment has the potential to corrupt valuable data required to determine the circumstances of an accident.

¹² These safety issues identify lessons to be learned. They do not merit a safety recommendation based on this investigation alone. However, they may be used for analysing trends in marine accidents or in support of a future safety recommendation